Benchmark progress: crypto-autoresearcher

From bench.record/v1 records in Git; raw data in S3 by SHA-256. Derived numbers are labelled derived.

Records: open 0, measured 14, invalid 2, censored 0. Headline unit S: group operations / sqrt(n); the ECDLP cost unit, every phase priced.

Series in unit bytes: gain vs first record (derived, %)

Series in unit count: gain vs first record (derived, %)

Series in unit fmul: gain vs first record (derived, %)

Series summary

problem • metric	unit	first	latest	gain vs first	value/ref	pts
ic:prime-field-cold-solvers • cancellation_first	fmul	813,356	813,356	?	?	1
ic:prime-field-cold-solvers • cancellation_first	count	506,680	506,680	?	?	1
ic:prime-field-cold-solvers • full_joint_fr_mul	fmul	1.617e+06	1.617e+06	?	?	1
ic:prime-field-cold-solvers • full_joint_fr_sub	count	537,790	537,790	?	?	1
ic:prime-field-cold-solvers • matched_target_out	count	288	288	?	?	1
ic:prime-field-fresh-holdouts • distinct_targets	count	8	8	?	?	1
ic:prime-field-fresh-holdouts • distinct_targets	count	15	15	?	?	1
ic:prime-field-fresh-holdouts • source_construct	fmul	98,566	98,566	?	?	1
ic:prime-field-fresh-holdouts • source_queries	count	49,152	49,152	?	?	1
ic:prime-field-source-budgets • distinct_targets	count	10	10	?	?	1
ic:prime-field-source-budgets • first_checked_in	count	96	96	?	?	1
ic:prime-field-source-budgets • first_checked_jo	count	48	48	?	?	1
ic:prime-field-source-budgets • source_queries	count	98,304	98,304	?	?	1
ic:prime-field-source-control • distinct_targets	count	8	8	?	?	1
ic:prime-field-source-control • source_queries	count	49,152	49,152	?	?	1
ic:prime-field-sparse-functionals • checked_solv	count	1,152	1,152	?	?	1
ic:prime-field-sparse-functionals • cumulative_o	count	1,156	1,156	?	?	1
ic:prime-field-sparse-functionals • sparse_cance	fmul	605,078	605,078	?	?	1
ic:prime-field-sparse-functionals • sparse_cance	count	2	2	?	?	1
ic:prime-field-sparse-functionals • sparse_cance	count	50	50	?	?	1
ic:prime-field-sparse-functionals • sparse_joint	fmul	688,054	688,054	?	?	1
kernel:prime-field-affine-target-holdout • affin	count	39,341	39,341	?	?	1
kernel:prime-field-affine-target-holdout • fresh	count	4	4	?	?	1
kernel:prime-field-affine-target-holdout • fresh	count	32	32	?	?	1
kernel:prime-field-affine-target-holdout • max_r	bytes	1.607e+08	1.607e+08	?	?	1
kernel:prime-field-affine-target-holdout • scope	count	1,152	1,152	?	?	1
kernel:prime-field-affine-target-holdout • solve	count	2,304	2,304	?	?	1
kernel:prime-field-affine-target-holdout • sourc	count	147,456	147,456	?	?	1
kernel:prime-field-affine-target-holdout • spars	count	602,296	602,296	?	?	1
kernel:prime-field-affine-target-holdout • spars	count	4,762	4,762	?	?	1
kernel:prime-field-affine-target-holdout • spars	fmul	6.851e+06	6.851e+06	?	?	1
kernel:prime-field-affine-target-holdout • spars	count	7.199e+06	7.199e+06	?	?	1
kernel:prime-field-affine-target-holdout • spars	count	456,714	456,714	?	?	1
kernel:prime-field-affine-target-holdout • spars	count	4,952	4,952	?	?	1
kernel:prime-field-affine-target-holdout • spars	fmul	6.415e+06	6.415e+06	?	?	1
kernel:prime-field-affine-target-holdout • spars	count	6.703e+06	6.703e+06	?	?	1
kernel:prime-field-affine-target-holdout • targe	count	33,728	33,728	?	?	1
kernel:prime-field-ap-source-holdout • construct	count	7,599	7,599	?	?	1
kernel:prime-field-ap-source-holdout • fresh_cur	count	4	4	?	?	1
kernel:prime-field-ap-source-holdout • fresh_tar	count	32	32	?	?	1
kernel:prime-field-ap-source-holdout • max_rss_b	bytes	9.612e+07	9.612e+07	?	?	1
kernel:prime-field-ap-source-holdout • scope_cas	count	432	432	?	?	1
kernel:prime-field-ap-source-holdout • solver_ca	count	864	864	?	?	1
kernel:prime-field-ap-source-holdout • source_qu	count	147,456	147,456	?	?	1
kernel:prime-field-ap-source-holdout • sparse_ca	count	35,965	35,965	?	?	1
kernel:prime-field-ap-source-holdout • sparse_ca	count	6,774	6,774	?	?	1
kernel:prime-field-ap-source-holdout • sparse_ca	fmul	764,255	764,255	?	?	1
kernel:prime-field-ap-source-holdout • sparse_ca	count	706,961	706,961	?	?	1
kernel:prime-field-ap-source-holdout • sparse_jo	count	35,920	35,920	?	?	1
kernel:prime-field-ap-source-holdout • sparse_jo	count	7,970	7,970	?	?	1
kernel:prime-field-ap-source-holdout • sparse_jo	fmul	786,587	786,587	?	?	1
kernel:prime-field-ap-source-holdout • sparse_jo	count	700,348	700,348	?	?	1
kernel:prime-field-ap-source-holdout • target_qu	count	1,536	1,536	?	?	1
kernel:prime-field-batched-target-holdout • max_	bytes	1.941e+08	1.941e+08	?	?	1
kernel:prime-field-batched-target-holdout • obse	count	4	4	?	?	1

kernel:prime-field-batched-target-holdout • obse	count	32	32	?	?	1
kernel:prime-field-batched-target-holdout • quer	count	120,800	120,800	?	?	1
kernel:prime-field-batched-target-holdout • quer	count	3,775	3,775	?	?	1
kernel:prime-field-batched-target-holdout • quer	fmul	1.076e+06	1.076e+06	?	?	1
kernel:prime-field-batched-target-holdout • quer	count	1.087e+06	1.087e+06	?	?	1
kernel:prime-field-batched-target-holdout • quer	count	0	0	?	?	1
kernel:prime-field-batched-target-holdout • quer	count	241,600	241,600	?	?	1
kernel:prime-field-batched-target-holdout • quer	fmul	724,800	724,800	?	?	1
kernel:prime-field-batched-target-holdout • quer	count	1.691e+06	1.691e+06	?	?	1
kernel:prime-field-batched-target-holdout • sour	count	196,608	196,608	?	?	1
kernel:prime-field-batched-target-holdout • spar	count	31,177	31,177	?	?	1
kernel:prime-field-batched-target-holdout • spar	count	1,869	1,869	?	?	1
kernel:prime-field-batched-target-holdout • spar	fmul	347,426	347,426	?	?	1
kernel:prime-field-batched-target-holdout • spar	count	386,075	386,075	?	?	1
kernel:prime-field-batched-target-holdout • spar	count	4,245	4,245	?	?	1
kernel:prime-field-batched-target-holdout • spar	count	1,384	1,384	?	?	1
kernel:prime-field-batched-target-holdout • spar	fmul	311,016	311,016	?	?	1
kernel:prime-field-batched-target-holdout • spar	count	351,255	351,255	?	?	1
kernel:prime-field-batched-target-holdout • targ	count	44,992	44,992	?	?	1
kernel:prime-field-batched-target-holdout • veri	count	2,304	2,304	?	?	1
kernel:prime-field-batched-target-holdout • veri	count	1,152	1,152	?	?	1
kernel:prime-field-mixed-known-helpers • curves	count	4	4	?	?	1
kernel:prime-field-mixed-known-helpers • hybrid_	count	126,837	126,837	?	?	1
kernel:prime-field-mixed-known-helpers • hybrid_	count	913	913	?	?	1
kernel:prime-field-mixed-known-helpers • hybrid_	fmul	736,257	736,257	?	?	1
kernel:prime-field-mixed-known-helpers • hybrid_	count	972,666	972,666	?	?	1
kernel:prime-field-mixed-known-helpers • hybrid_	count	126,837	126,837	?	?	1
kernel:prime-field-mixed-known-helpers • hybrid_	count	913	913	?	?	1
kernel:prime-field-mixed-known-helpers • hybrid_	fmul	1.132e+06	1.132e+06	?	?	1
kernel:prime-field-mixed-known-helpers • hybrid_	count	1.361e+06	1.361e+06	?	?	1
kernel:prime-field-mixed-known-helpers • known32	count	1,138	1,138	?	?	1
kernel:prime-field-mixed-known-helpers • known32	count	27	27	?	?	1
kernel:prime-field-mixed-known-helpers • known32	fmul	1,801	1,801	?	?	1
kernel:prime-field-mixed-known-helpers • known32	count	7,672	7,672	?	?	1
kernel:prime-field-mixed-known-helpers • known32	count	1,138	1,138	?	?	1
kernel:prime-field-mixed-known-helpers • known32	count	27	27	?	?	1
kernel:prime-field-mixed-known-helpers • known32	fmul	8,221	8,221	?	?	1
kernel:prime-field-mixed-known-helpers • known32	count	14,947	14,947	?	?	1
kernel:prime-field-mixed-known-helpers • known48	count	2,271	2,271	?	?	1
kernel:prime-field-mixed-known-helpers • known48	count	35	35	?	?	1
kernel:prime-field-mixed-known-helpers • known48	fmul	4,083	4,083	?	?	1
kernel:prime-field-mixed-known-helpers • known48	count	21,944	21,944	?	?	1
kernel:prime-field-mixed-known-helpers • known48	count	2,271	2,271	?	?	1
kernel:prime-field-mixed-known-helpers • known48	count	35	35	?	?	1
kernel:prime-field-mixed-known-helpers • known48	fmul	17,247	17,247	?	?	1
kernel:prime-field-mixed-known-helpers • known48	count	40,116	40,116	?	?	1
kernel:prime-field-mixed-known-helpers • max_rss	bytes	2.513e+08	2.513e+08	?	?	1
kernel:prime-field-mixed-known-helpers • norm_pa	count	11,150	11,150	?	?	1
kernel:prime-field-mixed-known-helpers • norm_pa	count	991	991	?	?	1
kernel:prime-field-mixed-known-helpers • norm_pa	fmul	232,164	232,164	?	?	1
kernel:prime-field-mixed-known-helpers • norm_pa	count	251,826	251,826	?	?	1
kernel:prime-field-mixed-known-helpers • norm_sp	count	11,150	11,150	?	?	1
kernel:prime-field-mixed-known-helpers • norm_sp	count	991	991	?	?	1
kernel:prime-field-mixed-known-helpers • norm_sp	fmul	236,836	236,836	?	?	1
kernel:prime-field-mixed-known-helpers • norm_sp	count	256,067	256,067	?	?	1
kernel:prime-field-mixed-known-helpers • public_	count	32	32	?	?	1
kernel:prime-field-mixed-known-helpers • query_b	count	358,404	358,404	?	?	1
kernel:prime-field-mixed-known-helpers • query_b	count	8,962	8,962	?	?	1

kernel:prime-field-mixed-known-helpers • query_b	fmul	3.212e+06	3.212e+06	?	?	1
kernel:prime-field-mixed-known-helpers • query_b	count	3.226e+06	3.226e+06	?	?	1
kernel:prime-field-mixed-known-helpers • query_s	count	4	4	?	?	1
kernel:prime-field-mixed-known-helpers • query_s	count	716,798	716,798	?	?	1
kernel:prime-field-mixed-known-helpers • query_s	fmul	2.15e+06	2.15e+06	?	?	1
kernel:prime-field-mixed-known-helpers • query_s	count	5.018e+06	5.018e+06	?	?	1
kernel:prime-field-mixed-known-helpers • signed_	count	491,520	491,520	?	?	1
kernel:prime-field-mixed-known-helpers • signed_	count	225,280	225,280	?	?	1
kernel:prime-field-mixed-known-helpers • verifie	count	576	576	?	?	1
kernel:prime-field-mixed-known-helpers • verifie	count	2,496	2,496	?	?	1
kernel:prime-field-mixed-known-helpers • verifie	count	1,248	1,248	?	?	1
kernel:prime-field-paid-known-base-control • kno	count	1,211	1,211	?	?	1
kernel:prime-field-paid-known-base-control • kno	count	38	38	?	?	1
kernel:prime-field-paid-known-base-control • kno	fmul	1,940	1,940	?	?	1
kernel:prime-field-paid-known-base-control • kno	count	8,229	8,229	?	?	1
kernel:prime-field-paid-known-base-control • kno	count	1,211	1,211	?	?	1
kernel:prime-field-paid-known-base-control • kno	count	38	38	?	?	1
kernel:prime-field-paid-known-base-control • kno	fmul	8,270	8,270	?	?	1
kernel:prime-field-paid-known-base-control • kno	count	15,839	15,839	?	?	1
kernel:prime-field-paid-known-base-control • max	bytes	1.717e+08	1.717e+08	?	?	1
kernel:prime-field-paid-known-base-control • nor	count	11,355	11,355	?	?	1
kernel:prime-field-paid-known-base-control • nor	count	806	806	?	?	1
kernel:prime-field-paid-known-base-control • nor	fmul	133,623	133,623	?	?	1
kernel:prime-field-paid-known-base-control • nor	count	160,034	160,034	?	?	1
kernel:prime-field-paid-known-base-control • nor	count	1,835	1,835	?	?	1
kernel:prime-field-paid-known-base-control • nor	count	586	586	?	?	1
kernel:prime-field-paid-known-base-control • nor	fmul	115,183	115,183	?	?	1
kernel:prime-field-paid-known-base-control • nor	count	140,855	140,855	?	?	1
kernel:prime-field-paid-known-base-control • obs	count	4	4	?	?	1
kernel:prime-field-paid-known-base-control • obs	count	32	32	?	?	1
kernel:prime-field-paid-known-base-control • que	count	143,360	143,360	?	?	1
kernel:prime-field-paid-known-base-control • que	count	4,480	4,480	?	?	1
kernel:prime-field-paid-known-base-control • que	fmul	1.277e+06	1.277e+06	?	?	1
kernel:prime-field-paid-known-base-control • que	count	1.29e+06	1.29e+06	?	?	1
kernel:prime-field-paid-known-base-control • que	count	0	0	?	?	1
kernel:prime-field-paid-known-base-control • que	count	286,720	286,720	?	?	1
kernel:prime-field-paid-known-base-control • que	fmul	860,160	860,160	?	?	1
kernel:prime-field-paid-known-base-control • que	count	2.007e+06	2.007e+06	?	?	1
kernel:prime-field-paid-known-base-control • sou	count	196,608	196,608	?	?	1
kernel:prime-field-paid-known-base-control • tar	count	90,112	90,112	?	?	1
kernel:prime-field-paid-known-base-control • ver	count	1,248	1,248	?	?	1
kernel:prime-field-paid-known-base-control • ver	count	624	624	?	?	1
kernel:prime-field-signed-pair-construction • ba	count	12,992	12,992	?	?	1
kernel:prime-field-signed-pair-construction • ba	count	16	16	?	?	1
kernel:prime-field-signed-pair-construction • ba	fmul	117,024	117,024	?	?	1
kernel:prime-field-signed-pair-construction • ba	count	156,544	156,544	?	?	1
kernel:prime-field-signed-pair-construction • ba	bytes	2.942e+06	2.942e+06	?	?	1
kernel:prime-field-signed-pair-construction • ma	bytes	1.559e+08	1.559e+08	?	?	1
kernel:prime-field-signed-pair-construction • ne	count	0	0	?	?	1
kernel:prime-field-signed-pair-construction • sc	count	0	0	?	?	1
kernel:prime-field-signed-pair-construction • sc	count	51,968	51,968	?	?	1
kernel:prime-field-signed-pair-construction • sc	fmul	155,904	155,904	?	?	1
kernel:prime-field-signed-pair-construction • sc	count	364,416	364,416	?	?	1
kernel:prime-field-signed-pair-construction • sc	bytes	0	0	?	?	1
kernel:prime-field-signed-pair-construction • ve	count	32	32	?	?	1
kernel:prime-field-signed-pair-construction • ve	count	17,920	17,920	?	?	1
kernel:prime-field-target-effort • curves	count	4	4	?	?	1
kernel:prime-field-target-effort • hybrid8_parti	count	115,978	115,978	?	?	1

kernel:prime-field-target-effort • hybrid8_parti	count	860	860	?	?	1
kernel:prime-field-target-effort • hybrid8_parti	fmul	694,644	694,644	?	?	1
kernel:prime-field-target-effort • hybrid8_parti	count	925,733	925,733	?	?	1
kernel:prime-field-target-effort • hybrid8_spars	count	115,978	115,978	?	?	1
kernel:prime-field-target-effort • hybrid8_spars	count	860	860	?	?	1
kernel:prime-field-target-effort • hybrid8_spars	fmul	1.076e+06	1.076e+06	?	?	1
kernel:prime-field-target-effort • hybrid8_spars	count	1.302e+06	1.302e+06	?	?	1
kernel:prime-field-target-effort • known32_8_par	count	1,260	1,260	?	?	1
kernel:prime-field-target-effort • known32_8_par	count	29	29	?	?	1
kernel:prime-field-target-effort • known32_8_par	fmul	1,896	1,896	?	?	1
kernel:prime-field-target-effort • known32_8_par	count	8,241	8,241	?	?	1
kernel:prime-field-target-effort • known32_8_spa	count	1,260	1,260	?	?	1
kernel:prime-field-target-effort • known32_8_spa	count	29	29	?	?	1
kernel:prime-field-target-effort • known32_8_spa	fmul	8,238	8,238	?	?	1
kernel:prime-field-target-effort • known32_8_spa	count	16,004	16,004	?	?	1
kernel:prime-field-target-effort • known48_24_pa	count	1,017	1,017	?	?	1
kernel:prime-field-target-effort • known48_24_pa	count	19	19	?	?	1
kernel:prime-field-target-effort • known48_24_pa	fmul	4,245	4,245	?	?	1
kernel:prime-field-target-effort • known48_24_pa	count	21,390	21,390	?	?	1
kernel:prime-field-target-effort • known48_24_sp	count	1,017	1,017	?	?	1
kernel:prime-field-target-effort • known48_24_sp	count	19	19	?	?	1
kernel:prime-field-target-effort • known48_24_sp	fmul	17,421	17,421	?	?	1
kernel:prime-field-target-effort • known48_24_sp	count	34,128	34,128	?	?	1
kernel:prime-field-target-effort • known48_8_par	count	2,781	2,781	?	?	1
kernel:prime-field-target-effort • known48_8_par	count	42	42	?	?	1
kernel:prime-field-target-effort • known48_8_par	fmul	4,818	4,818	?	?	1
kernel:prime-field-target-effort • known48_8_par	count	26,060	26,060	?	?	1
kernel:prime-field-target-effort • known48_8_spa	count	2,781	2,781	?	?	1
kernel:prime-field-target-effort • known48_8_spa	count	42	42	?	?	1
kernel:prime-field-target-effort • known48_8_spa	fmul	18,672	18,672	?	?	1
kernel:prime-field-target-effort • known48_8_spa	count	47,104	47,104	?	?	1
kernel:prime-field-target-effort • max_rss_bytes	bytes	3.898e+08	3.898e+08	?	?	1
kernel:prime-field-target-effort • norm8_partial	count	191	191	?	?	1
kernel:prime-field-target-effort • norm8_partial	count	711	711	?	?	1
kernel:prime-field-target-effort • norm8_partial	fmul	148,935	148,935	?	?	1
kernel:prime-field-target-effort • norm8_partial	count	174,473	174,473	?	?	1
kernel:prime-field-target-effort • norm8_sparse_	count	191	191	?	?	1
kernel:prime-field-target-effort • norm8_sparse_	count	711	711	?	?	1
kernel:prime-field-target-effort • norm8_sparse_	fmul	153,645	153,645	?	?	1
kernel:prime-field-target-effort • norm8_sparse_	count	178,490	178,490	?	?	1
kernel:prime-field-target-effort • public_target	count	32	32	?	?	1
kernel:prime-field-target-effort • query_batch_f	count	465,664	465,664	?	?	1
kernel:prime-field-target-effort • query_batch_f	count	11,194	11,194	?	?	1
kernel:prime-field-target-effort • query_batch_f	fmul	4.178e+06	4.178e+06	?	?	1
kernel:prime-field-target-effort • query_batch_f	count	4.191e+06	4.191e+06	?	?	1
kernel:prime-field-target-effort • query_scalar_	count	0	0	?	?	1
kernel:prime-field-target-effort • query_scalar_	count	931,328	931,328	?	?	1
kernel:prime-field-target-effort • query_scalar_	fmul	2.794e+06	2.794e+06	?	?	1
kernel:prime-field-target-effort • query_scalar_	count	6.519e+06	6.519e+06	?	?	1
kernel:prime-field-target-effort • signed_source	count	491,520	491,520	?	?	1
kernel:prime-field-target-effort • signed_target	count	439,808	439,808	?	?	1
kernel:prime-field-target-effort • verified_info	count	576	576	?	?	1
kernel:prime-field-target-effort • verified_solv	count	2,520	2,520	?	?	1
kernel:prime-field-target-effort • verified_syst	count	1,260	1,260	?	?	1
prime-field-factor-bases • actual_batch_calls	count	12	12	?	?	2
prime-field-factor-bases • actual_online_engines	count	4	4	?	?	2
prime-field-factor-bases • candidate_geometric_q	count	46	183	?	?	2
prime-field-factor-bases • candidate_skipped_log	count	722	585	?	?	2

prime-field-factor-bases • candidate_targets	count	32	29	?	?	2
prime-field-factor-bases • construction_only_tar	count	27	27	?	?	1
prime-field-factor-bases • construction_target_c	count	8	8	?	?	1
prime-field-factor-bases • extra_new_target_cert	count	5	12	?	?	2
prime-field-factor-bases • full_cold_batch_fp_fi	count	15,409	16,543	?	?	2
prime-field-factor-bases • full_cold_batch_fp_fi	count	434,337	439,353	?	?	2
prime-field-factor-bases • full_cold_batch_fr_fi	count	32	25	?	?	2
prime-field-factor-bases • full_cold_batch_fr_fi	count	2,441	886	?	?	2
prime-field-factor-bases • full_cold_batch_pair_	bytes	1.024e+06	1.024e+06	?	?	2
prime-field-factor-bases • full_cold_batch_query	bytes	6.271e+06	6.271e+06	?	?	2
prime-field-factor-bases • natural_new_target_ce	count	0	9	?	?	2
prime-field-factor-bases • omitted_reference_equ	count	274	59	?	?	2
prime-field-factor-bases • reference_geometric_q	count	768	768	?	?	2
prime-field-factor-bases • reference_targets	count	32	29	?	?	2
prime-field-factor-bases • retained_nonrecoverie	count	3	3	?	?	1
prime-field-factor-bases • suppress_cold_batch_f	count	14,139	15,840	?	?	2
prime-field-factor-bases • suppress_cold_batch_f	count	120,789	186,279	?	?	2
prime-field-factor-bases • suppress_cold_batch_f	count	32	25	?	?	2
prime-field-factor-bases • suppress_cold_batch_f	count	1,010	711	?	?	2
prime-field-factor-bases • suppress_cold_batch_p	bytes	1.024e+06	1.024e+06	?	?	2
prime-field-factor-bases • suppress_cold_batch_q	bytes	379,968	1.498e+06	?	?	2
prime-field-factor-bases • verified_native_cases	count	8	8	?	?	2

Every value is quoted from the record it names, in that record’s unit. Percentages and ratios under ’derived’ are computed here from those values and are not measurements. Series in different units are never compared.